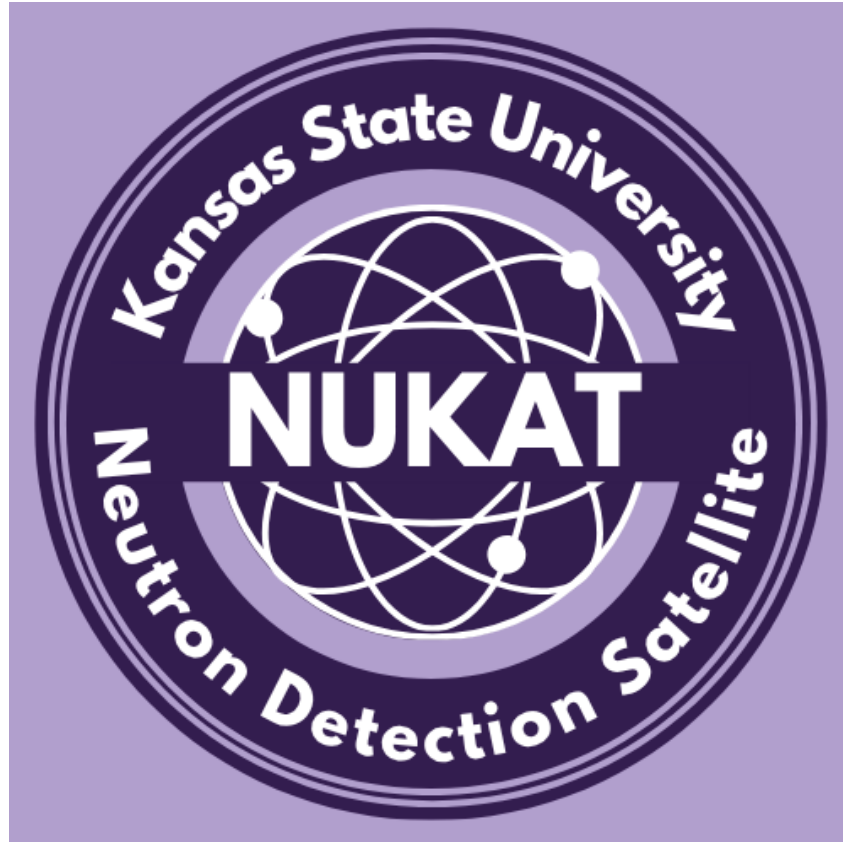




## **Neutron Detector: K-State Analysis Tool (NuKAT) CubeSat Mission**

Payload Provider / Sponsor: Kansas State University

Faculty Advisor / Instructor: Paul Snider – [pmsnider@ksu.edu](mailto:pmsnider@ksu.edu) – (785) 532-5590

Principal Investigator: Colby Johnston – [cubesat@k-state.edu](mailto:cubesat@k-state.edu) – (913) 800-1676

Chief Engineer: Bailey Walke – [walke1@ksu.edu](mailto:walke1@ksu.edu) – (515) 865 - 0063

## Executive Summary

The NuKAT CubeSat mission is a 2U CubeSat developed by Kansas State University for a technology demonstration of the Dry Storage Cannister (DSC) Domino neutron detector platform developed by Radiation Detection Technologies, Inc. (RDT). The primary objective is to collect location-specific Low Earth Orbit neutron flux data across four energy bins (thermal, epithermal, and two fast regions), specifically the South Atlantic Anomaly and polar regions. This data will be distributed to academic and research institutions to support education and research on the space radiation environment and new detector technologies.

Other mission objectives are to flight qualify the detector platform in space, build flight heritage for the DSC-Domino and the CubeSat bus for future missions and to provide educational benefits for both the students developing and operating the satellite, and other stakeholders benefiting from the scientific data the mission provides.

## Science

The mission uses four DSC-Domino platforms (roughly  $67 \times 27 \times 4$  mm each) to demonstrate the technology of a science payload in a 2U CubeSat. Two platforms are encased in a high-density polyethylene (HDPE) moderator to shift their neutron energy relative efficiency peak to the epithermal region (100 eV). Two platforms are unmoderated with relative efficiency peaking in the thermal (0.0253 eV) and fast (18 MeV) energy regimes. The doubling of platforms in moderated and unmoderated configurations provides redundancy for the key science components. The moderator is an extruded stadium geometry with  $33.8 \times 86.8 \times 45.9$  mm outer dimensions and fits in 1U of a CubeSat, allowing the other 1U to house the unmoderated platforms along with all other necessary satellite systems. Thus, this payload is housed in a 2U satellite and can fit into a 3U or 6U dispenser.

The DSC-Domino detector platform is at technology readiness level (TRL) 6 (as defined by NASA) because of the space qualification of the micro-structured semiconductor neutron detector (MSND) component on the original Domino platform as part of the Department of Defense Space Test Program's NeRDI-1A mission. Therefore, the underlying MSND technology is flight qualified and the importance of this mission comes from the DSC-Domino's ability to gather both thermal and fast neutron counts on a single platform, along with simpler communication due to its integrated driver board. Thus, this mission improves the TRL of the DSC-Domino to a minimum of TRL 7 with a TRL 8 ceiling. The platform utilizes four MSNDs to measure background-corrected fast and thermal neutrons in a low-profile, low-mass, low-power, and low-cost package as an alternative to existing, similarly capable detector systems.

Beyond the technological value for other missions, the payload records four energy bins with associated positional and temporal relations, providing beneficial data to universities and research institutions, especially across regions that are often neglected due to their high radiation flux (specifically the South Atlantic Anomaly and polar regions). This results in better radiation environment models, records of the effects from space anomalies (such as solar flares), and classroom examples/laboratory exercises for developing an understanding and intuition of the MSND platform.

## Spacecraft

The CubeSat is assembled by K-State students, with most bus components being commercial off-the-shelf (COTS) with flight heritage -except the external thermal paneling, aluminum chassis, and the deployment arm for the magnetometer, all of which are manufactured in collaboration with K-State's Technology Development Institute (TDI). Component manufacturers include EnduroSat, DHV, NextAge, RDT, TensorTech, K-State TDI, and McMaster Carr. The stowed and deployed configurations for the payload are shown in Figure 1.

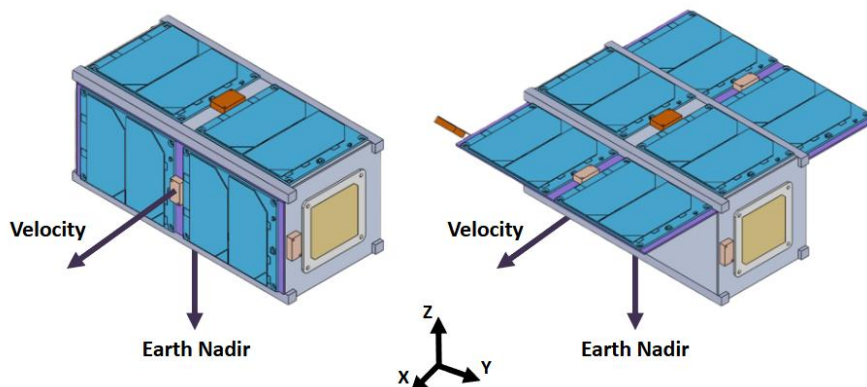


Figure 1: NuKat Stowed and Deployed

The Spacecraft does not possess any propulsion or pressurized systems; thus, the dry and wet masses are the same. The deployable mass includes a burn wire system for the UHF antenna, which will be between 30g and 210g. The magnetometer has a mass of about 16 g with a deployment system mass of about 195 g. The estimated total satellite mass is 2.5kg.

The Spacecraft uses a rail-type dispenser along the X-axis and can be placed anywhere in a 3U or larger dispenser for launch vehicle deployment. The overall maximum volume of the Spacecraft is within the specifications of the standard dispenser, with overall dimensions of 120 mm Y × 120 mm Z × 227 mm X. There are no special sensitivities or accommodations for launch.

The Spacecraft operates in a sun-synchronous circular orbit at an altitude of approximately 500 km with an inclination of approximately 97 degrees. The orbit altitude can extend up to 590 km with added attitude maneuvers to decrease lifetime. The right ascension of the ascending node, RAAN, for the Spacecraft to pass over K-State's Manhattan campus is about 250 degrees.

The payload end use, beyond qualifying the DSC-Domino in space, is to collect and distribute neutron count data to academic and research institutions for educational and research purposes.

## Do No Harm

The Spacecraft does not contain any pressurized systems, propulsion systems, or toxic propellants. The primary hazards for the Spacecraft are the Lithium-Ion (Li-ion) battery system, which poses a risk of thermal runaway and fire, and the deployable solar arrays and magnetometer, which must be securely stowed to prevent premature release during launch. According to the United States Department of Transportation, the Li-ion battery is a Class 9 hazardous material, which requires specialized packaging and documentation for shipping, that we will provide to ensure compliance.

The Electrical Power & Distribution Subsystem (EPDS) has redundant hardware and software protection against battery overcharge and deep discharge, verified by rigorous cycle and abuse testing. The deployable arrays are secured by a non-flammable retention mechanism with a safety margin of at least twice the maximum launch load, verified by full-scale vibration and shock testing.

The Spacecraft will be powered off from the time it is shipped to the launch site, up through the deployment from the launch vehicle using a remove-before-flight power inhibit plug that will physically ensure the Spacecraft is not powered. There will be an initialization phase shortly after launch vehicle deployment that will charge, power on, and deploy appendages.

The Spacecraft will undergo comprehensive structural, electrical, and thermal testing to verify integrity under launch conditions, with certain tests conducted in-house and outsourcing others. Additionally, all components are designed to ensure outgassing below NASA's total mass loss coefficient of 1%. The assembly, integration, and testing (AIT) facilities are required to ensure all hardware is "Visually Clean" under NASA's designation and to verify that no debris is present.

## Organization

Our team has 23 engineering students from both the Manhattan and Salina campuses, representing the following engineering majors: Mechanical, Nuclear, Industrial, Electrical, Computer Science, and Engineering Technology and Uncrewed Aircraft Systems (UAS) as shown in Figure 2. Most students are undergraduate seniors or in graduate school, so the team is comprised of people who already have experience with technical work and team collaboration.

The advisors for the project are Mr. Paul Snider and Dr. Raju Dandu from Kansas State University. Additional guidance comes from a mission advisory board comprised of Kansas State University faculty and active and retired aerospace employees. Kansas State University has an amateur radio club W0QQQ on our Manhattan, KS campus that will provide command and control up-link and data downlink for the mission.

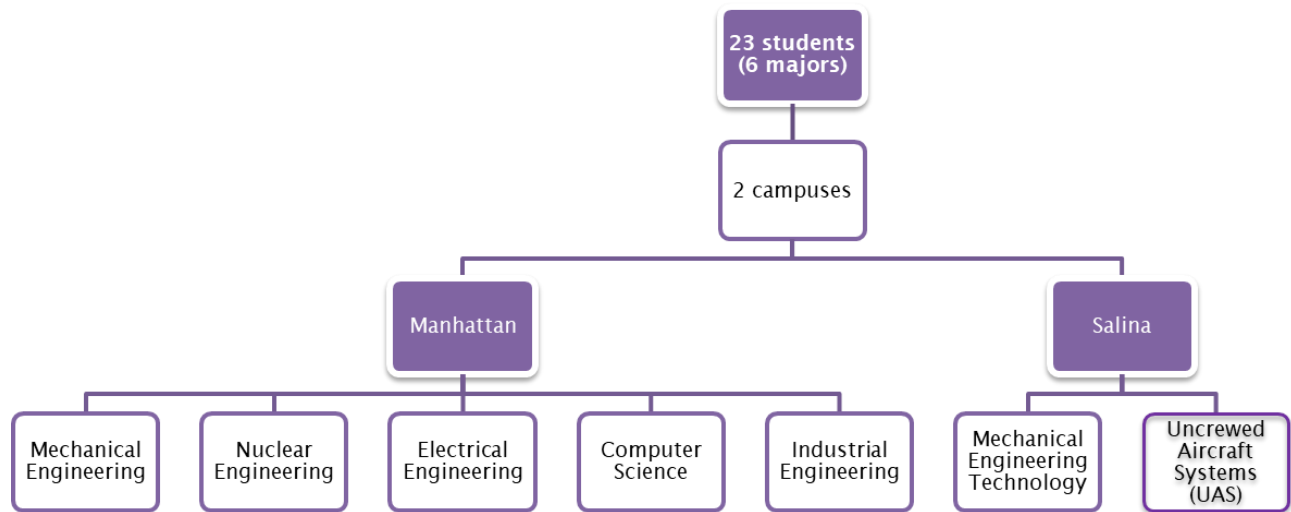
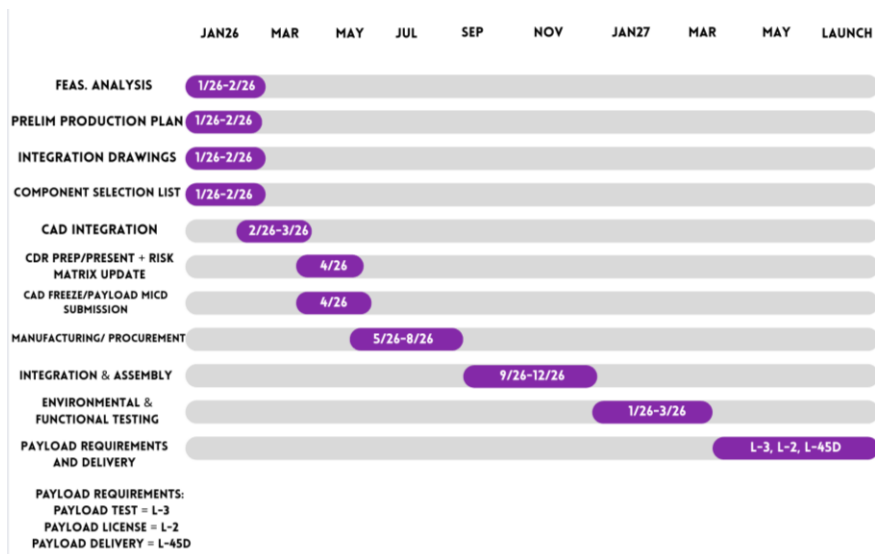


Figure 2: Organization Chart

## Facility

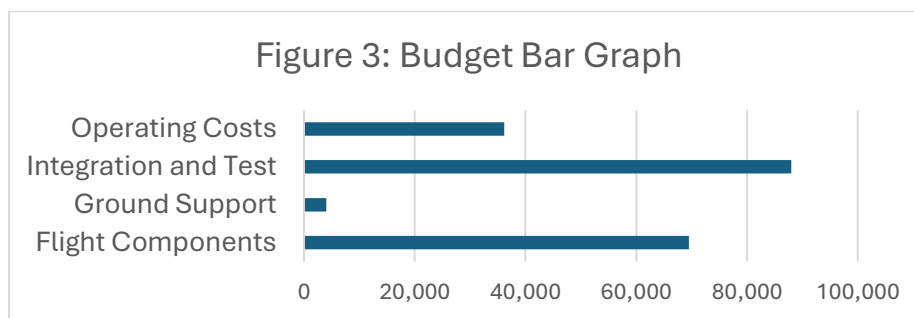
The Integration and Test Facility (ITF) serves to support the assembly, integration, and testing of the satellite and its subsystems. The ITF provides a secure space on the Kansas State University Manhattan campus with all the necessary equipment to carry out assembly operations and integration and testing procedures. For satellite electrical systems, ITF is equipped with instruments to assemble, measure, and stimulate/simulate sensors and actuators. Ground Support Equipment includes high-resolution measurement instruments and precise assembly tools. We will outsource the mechanical and thermal testing capabilities to ensure the satellite's structures and components can endure the vibration and thermal stresses that are standard in launch and space environments.

## Schedule



## Budget

The current best estimate for the NuKAT mission with a 20% margin is \$197,000, not including launch costs, which is allocated to the categories shown in Figure 3. Operating Costs cover mission operations, travel, and miscellaneous expenses. Integration and Test include in-house and outsourced testing, plus launch and payload integration. Ground Support consists of ground station hardware and software. Flight Components encompass all satellite parts necessary for assembly and operation in space, including structure, subsystems, and flight software.



Kansas State University has received a \$50,000 gift from a donor for our CubeSat efforts. The remaining funds for the mission will be raised through fundraising efforts, including outreach to the KSU Foundation and exploring sponsorship and in-kind gift opportunities with our suppliers.

## Regulatory

The mission will be conducted in full compliance with all applicable national and international regulatory requirements for the launch, deployment, and operation of the payload. We have identified the following licenses/authorizations as required for this mission: Federal Communications Commission (FCC), International Amateur Radio Union (IARU), Federal Aviation Administration (FAA), International Telecommunication Union (ITU), and additional licenses that may be required throughout the process. We currently have not applied for any licenses for the mission.